1-Bit ALU Design Project

CMOS Layout, Simulation and Parasitic Analysis

Justice J. Rebrovich

Dr. Sudhanshu Choudhary

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2µm CMOS Technology • VDD = 5V • L-Edit + LTSpice

1. Introduction

This report presents the design, layout, simulation and parasitic analysis of a 1-bit Arithmetic Logic Unit (ALU) implemented in 2µm CMOS technology. The ALU supports four operations selected by a 2-bit control signal using three cascaded 2:1 multiplexer stages.

All gate cells were drawn from transistor level in L-Edit using lambda-based design rules, verified with Design Rule Check (DRC), simulated in LTSpice for ideal behavior, and compared against extracted parasitic netlists to quantify the impact of layout parasitics on propagation delay.

2. ALU Architecture

2.1 Operations

The 1-bit ALU computes four operations based on select inputs S1 and S0:

S1	S0	Operation
0	0	AND (A · B)
0	1	OR (A + B)
1	0	XOR (A ⊕ B)
1	1	ADD (A + B + Cin)

2.2 Gate-Level Implementation

The ALU is built from the following gate cells, all implemented at transistor level:

- AND gate: NAND3 + INV (used inside the MUX AND3 path)
- XOR gate: 4 × NAND2 (used in Full Adder SUM path)
- NOR gate: NOR2 + INV (output OR stage)
- Full Adder: 2 × XOR + carry NAND/NOR logic
- 2:1 MUX: INV + 3 × NAND2

- 3 cascaded 2:1 MUX stages for final output selection

3. CMOS Layout in L-Edit

3.1 Technology and Design Rules

All cells were drawn in L-Edit using 2µm CMOS lambda-based design rules. The following rules were applied throughout:

Design Rule	Value	Note
Poly width	2λ	Gate minimum
Poly overhang past active	2λ	Top and bottom
Poly-to-poly spacing	2λ	Between strips
Active width (NAND3)	22λ	3 poly + contacts
Contact size	2λ × 2λ	Process fixed
Contact to poly spacing	2λ	Min clearance
Active enclosure of contact	2λ	All sides
Metal 1 minimum	4λ × 4λ	1λ enc each side
P-Select enclosure	2λ	Around PMOS active
N-Well enclosure of active	6λ	From DRC verification
N-Well to N-Well spacing	6λ	Minimum

3.2 PMOS and NMOS Transistor Dimensions

Each transistor was sized to the following minimum DRC-clean dimensions:

- Active width: 22λ (fits source contact + 3 poly strips + drain contact with all spacing rules)
- Active height: 6λ (2λ enclosure + 2λ contact + 2λ enclosure)
- N-Well: 34λ wide × 16λ tall (6λ enclosure past active on all sides)
- N-Select / P-Select: 26λ wide × 10λ tall (2λ enclosure past active)
- Metal 1: minimum 4λ × 4λ to legally enclose a 2λ × 2λ contact

3.3 NAND3 Structure

The NAND3 gate uses one continuous active rectangle per rail with 3 poly strips crossing it. This shared diffusion approach avoids separate active regions, reducing mid-node metal stubs:

- PMOS: 3 transistors in parallel (shared source at VDD, all drains tied to Out)
- NMOS: 3 transistors in series (source chain from GND through mid-nodes to Out)
- 4 contacts per active region — one in each region between and outside the poly strips

- Mid-node contacts on the NMOS side are floating — the shared diffusion handles the series connection

4. LTSpice Simulation

4.1 Netlist Structure

The ALU was simulated in LTSpice using a hierarchical netlist. Each gate was defined as a subcircuit built from NAND2 and INV primitives at transistor level:

- NAND2: MP1, MP2 (PMOS parallel) + MN1, MN2 (NMOS series)
- INV: MP1 (PMOS) + MN1 (NMOS)
- 2:1 MUX: INV(SEL) + NAND(I0, SEL') + NAND(I1, SEL) + NAND(Na, Nb)
- Full Adder: 2 × XOR_GATE + 3 × NAND carry logic

Input signals used PULSE sources with staggered periods to exercise all 32 input combinations across the 640ns simulation window.

4.2 Ideal Delay Calculations (No Parasitics)

Theoretical propagation delays were calculated from first principles using the following parameters:

- VDD = 5V, VTn = VTp = 0.7V, KPn = 50µA/V², KPp = 25µA/V²
- NMOS: W = 2µm, L = 2µm | PMOS: W = 4µm, L = 2µm
- IDn = IDp = 462µA (matched drive currents)
- CL = 21.6 fF (gate capacitance per stage)

Operation	Ideal Delay (LTSpice)	Rise / Fall Time
AND	2221.6 ps	102.9 ps
OR	2192.4 ps	102.9 ps
XOR	2630.9 ps	102.9 ps
SUM (ADD)	3887.9 ps	102.9 ps

5. Parasitic Extraction and Analysis

5.1 Extracted Capacitance Values

Parasitic capacitances were extracted from the L-Edit layout. Key values from the extracted netlists:

Cell	Avg Node Cap	Total Cap
MUX	24.07 fF	288.8 fF
AND / NOR / XOR	40.72 fF	570.0 fF

Full Adder	—	746.4 fF
Full Adder max node	364.3 fF	—

5.2 Parasitic Delay Results

Extracted transistor dimensions from the netlist: PMOS W = 6µm, NMOS W = 6µm (larger than ideal sizing). This increased drive current to IDn = 1386.8µA, IDp = 693.4µA, partially compensating for the larger parasitic capacitances.

Measurement	LTSpice Ideal	L-Edit Parasitic	Δ Overhead	% Increase
tp AND	2221.6 ps	2968.2 ps	746.6 ps	**+33.6%**
tp OR	2192.4 ps	3073.3 ps	880.9 ps	**+40.2%**
tp XOR	2630.9 ps	2743.3 ps	112.4 ps	**+4.3%**
tp SUM	3887.9 ps	5544.7 ps	1656.8 ps	**+42.6%**
INV delay	116.9 ps	514.0 ps	397.1 ps	**+340%**
MUX delay	643.1 ps	774.3 ps	131.2 ps	**+20.4%**
Rise time	102.9 ps	603.0 ps	500.1 ps	**+490%**
Fall time	102.9 ps	301.5 ps	198.6 ps	**+193%**

5.3 Key Observations

- The SUM (ADD) path is the critical path at 5.54 ns with parasitics — 42.6% slower than ideal.
- XOR is the least affected operation (+4.3%) because the extracted transistors are wider (W=6µm vs W=2µm ideal), providing greater drive current that partially compensates for the larger CL.
- The Full Adder output node has the largest parasitic capacitance at 364.3 fF — approximately 17× larger than the ideal 21.6 fF used in hand calculations.
- INV delay is most severely affected (+340%) due to the 190 fF load capacitance on its output node.
- Rise time increases from 102.9 ps to 603 ps, a 5.9× increase driven by large output node capacitances in the extracted layout.
- MUX stages are least affected (+20.4%) as routing between stages is relatively short.

6. DRC Results

All cells achieved zero DRC errors after iterative correction. Key DRC violations encountered and resolved during layout:

- Source/Drain Active to Well Edge violation: N-Well initially too small. Fixed by expanding N-Well to provide 6λ enclosure past active on all sides (rule 2.3a).
- Poly overhang violations: Initial poly strips did not extend sufficiently past active. Corrected to 2λ overhang on all sides.
- Metal 1 enclosure violations: Metal rectangles initially 3λ wide. Corrected to minimum 4λ × 4λ to provide 1λ enclosure around 2λ × 2λ contacts.
- P-Select / N-Select overlap: Resolved by ensuring select layers do not overlap between PMOS and NMOS regions.

Final DRC status: 0 errors across all cells (INV, NAND3, NOR2, XOR, Full Adder, 2:1 MUX).

7. Conclusion

A 1-bit CMOS ALU was successfully designed at transistor level using L-Edit in 2µm technology. All four operations (AND, OR, XOR, ADD) were verified functionally in LTSpice. Parasitic extraction revealed significant performance degradation compared to ideal simulation, with the ADD path being the most critical at 5.54 ns — 42.6% slower than the ideal 3.89 ns.

The largest source of parasitic overhead was capacitance at the Full Adder output node (364.3 fF), emphasizing the importance of minimizing wire length and contact count at high-fanout nodes in CMOS layout. The XOR path benefited from larger extracted transistor widths (W=6µm) which partially compensated for increased load capacitance, limiting delay increase to only 4.3%.

It seemed like the biggest issue was making sure the poly to metal contacts avoided DRC errors. It was very picky with how it wanted it otherwise there seemed to be no issues.

The project demonstrated the complete VLSI design flow from gate-level architecture through transistor-level layout, DRC verification, ideal simulation, and parasitic comparison — a representative workflow for custom integrated circuit design.

References

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